Factory Resume

The day after half the population disappears: keeping essentials moving

In one line: Factory Resume rebuilds a fulfilment centre's plan every morning — who works where, which partners to bring in, which orders ship — so that essentials reach the people who are left. In the prototype's 5-day outlook it ships **100% of essential units on time vs. 59%** for business as usual, and **+37% more units** overall, for Rs 14 of partner spend per extra unit.

Why this problem, out of everything that breaks

When half the population vanishes overnight, hospitals, utilities and food supply all fail at once. I focused on **fulfilment — how essentials physically reach households** — for three reasons: every other recovery depends on it; its inputs (orders, rosters, stock) are already digital, so software can help within hours rather than months; and one manager's decisions reach thousands of homes. The closest real precedent is a pandemic lockdown, when e-commerce in several countries, India included, was briefly restricted to essential goods while warehouses ran short-staffed.

What actually breaks on day one

The intuitive framing — “half the staff, so half the capacity” — is wrong, because **demand drops too**. Half the customers are gone. The real problem is **mismatch**:

- **The wrong people are left.** Packing lost most of its team, picking lost the fewest. The line ships only as fast as its slowest step, and today nobody is deciding who should move.
- **Needs have shifted.** The customers who are left stockpile hand wash and sanitary pads, and stop buying serums. A first-due-first-shipped floor still ships the lip balm due today before the soap due tomorrow.
- **Help arrives late.** Partners need 6–48 hours to activate, and couriers are halved too.
- **Any plan is wrong by noon.** Who shows up, how fast people work in unfamiliar roles, and what stock lands all change daily.

Primary user: the operations manager of a fulfilment network (here, a fictional personal-care and hygiene brand with Bengaluru and Pune centres and 60 of ~120 staff left). They are accountable for what ships and must defend each call to leadership.

The solution: three structural ideas

Idea	What the engine does	Guardrail
Redeploy, don't just allocate	Each team has cross-skills (a picker can pack at 60% speed after 2h of ramp-up). The engine moves one worker at a time from the best-covered task to the bottleneck, but only when that lifts the weaker of the two, counting the speed and ramp-up loss.	No one is moved into certified work (quality inspection, inventory exceptions), and those teams are never moved out.
Priority from need, not a label	Each SKU is tagged essential, everyday or discretionary. Priority = need, then due date, with a one-line reason (“Essential · due in 1d”).	Tags are an operator policy input, never inferred by the engine.
A daily loop, not a one-off plan	Check in who showed up → plan → end-of-shift actuals → recalibrate productivity → tomorrow's plan. Backlog rolls over and re-ranks; new orders arrive at the shifted demand.	Partners count only for the hours they're live, and volume is bought only up to what the rest of the line can ship.

A morning with Factory Resume

- **7:00 · Check-in.** The manager enters who actually arrived. The plan rebuilds from that headcount, not the roster.
- **7:05 · Plan.** Packing is the bottleneck, and no packing partner can start until tomorrow. The engine moves 9 of 60 people across tasks. Day 1 ships 2,800 units vs. 2,200 with everyone on their own task, and 1,900 essential units vs. 0 under first-due-first.
- **7:10 · Request.** It flags QuickPack (24h activation) and ValleyWorks (48h) to request now, and explains why buying picking capacity today would be waste.
- **19:00 · Close.** Packing reports 80% of plan. Plan accuracy is logged, packing productivity is nudged to 90%, and tomorrow's plan starts from that.

MVP scope — and the tradeoffs behind it

- **An explainable heuristic over an optimal solver.** A manager won't stake their job on a black box; every move the engine makes fits in one sentence. The cost: the plan isn't guaranteed optimal.
- **A human approves; nothing books automatically.** Partner requests and redeployments are recommendations. Booking automatically saves minutes, but one wrong call costs trust.
- **Data comes from what teams already keep.** MVP input is the order export, the roster with cross-skills, and stock, not WMS integrations. *This is the riskiest assumption:* rosters may not record cross-skills today.
- **The daily loop ships in v1**, because re-planning when things get worse is part of day one, not an add-on. Live integrations, payroll and automatic booking are cut.

Success metrics

Type	Metric	Why this one
North star	Essential units shipped by their promised date	What the people who are left actually need; volume alone could be gamed by shipping lip balm.
Input	Time from check-in to approved plan; share of recommendations approved without edits	Speed matters only if the manager trusts the plan enough to act on it.
Plan accuracy	Shipped ÷ planned, per day	Measured by the loop itself. If it's low, the plan is fiction and the recalibration is the fix.
Efficiency	Partner spend per extra unit shipped	Makes outsourcing a priced decision before budget is committed.
Guardrails	Quality incidents in redeployed roles; partner over-commitment; delayed customers told before their date	Prevents winning the north star by cutting corners.

Prototype result, 5-day outlook. Same engine and same synthetic data. “Business as usual” = everyone on their own task, no partners, first-due-first-shipped.

	With the plan	Business as usual
Essential units on time (of 8,380 due)	100%	59%
Units shipped	15,120	11,000
Backlog left after Day 5	4,680	8,800
Partner spend	Rs 56,492	Rs 0

What the prototype taught me

The bottleneck moves. Packing binds on Day 1, picking on Day 2 once a packing partner is live, Pune's couriers on Days 3–4, and stock by Day 5. A plan made once on day one would be optimising the wrong constraint by Day 3, which is why the product is a loop. Next bets, in order: test whether managers can supply cross-skill data in under an hour; add courier re-routing between centres; and turn the “stock” constraint into replenishment requests.